

2. Analyze the two GenAI programs. The analysis must include:

a. Name of two GenAI programs

Claude and Gemini

b. How you accessed them (e.g., Google query, CoPilot on my computer)

Online off google for gemini, the app for claude.

c. Prompt used. (You must use the same prompt for both GenAIs.)

<the assignment file> followed by "The assignment is described within, generate the code accordingly. Try to handle incorrect input error."

d. Generated code from both GenAIs.

Gemini:

<https://drive.google.com/file/d/1K-tTlq1RWFT0UFSrHORNl69MRfGJV2VX/view?usp=sharing>

Claude:

<https://drive.google.com/file/d/1VoBWZ7OJIMPIREZF9PJPYfXjFkep7IPD/view?usp=sharing>

e. Analyze the code in terms of correctness, execution time, space complexity, and maintainability as explained in the "GenAI Code Comparison" lecture.

Correctness: both execute but Gemini's has worse error handling (for example, invalid dates could be entered).

Execution time/time complexity: since both implement a maxheap as the main data structure, time complexity is $O(\log(n))$, which is very efficient when scaled up.

Space Complexity: The email lists are both $O(n)$ but Claude has better allocation of memory so it performs better when really large amounts of data are used.

Maintainability: Both are very organized with their code structure and have comments on basically every line. Makes the code very easy to follow and make necessary changes to. Claude does however have robust function headers for each function, giving its purpose, and its inputs and outputs. This puts it easily ahead as far as maintainability goes.

f. Select one to use as the basis for this assignment and justify why you picked it.

Gemini simply has the most room for improvement across all areas.

g. Explain what you did to improve it in terms of correctness, execution time, space complexity, and maintainability

Some of the error handling was improved, as well as significantly more commentary and battery memory allocation. Overall the program should scale a lot better.

*The only collaboration was between myself and the two ai tools mentioned previously